Interactions

Cross section & interaction length

Probability $P(x)$ to propagate distance x without interaction:

$$P(x) = e^{-x/\lambda}$$

where λ is the interaction length:

$$\lambda = \frac{1}{n \cdot \sigma}$$

with n — number density of targets (units: cm^{-3})

$$n = \rho \cdot \frac{N_A}{A}$$

ρ = Target density
N_A = Avogadro number
A = molar mass

Example: Solar Neutrino